Program 8:
Deploy a Web Application to Kubernetes using Minikube
12/12/2025

Project Structure:

```
prog8
  |- nginx-configmap.yaml
  |- nginx-pods.yaml
  |- server.js
```

Steps for Execution:

Step 1: Check for prerequisites,

> minikube version
> kubectl version --client
> docker --version

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$
minikube version
kubectl version --client
docker --version
minikube version: v1.37.0
commit: 65318f4cfff9c12cc87ec9eb8f4cdd57b25047f3
Client Version: v1.31.0
Kustomize Version: v5.4.2
Docker version 29.0.4, build 3247a5a
```

Step 2: Create a project folder as prog7 and move into it

> mkdir prog8
> cd prog8/

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~$ cd devops_lab/
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ mkdir prog8
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ cd prog8/
```

Step 3: Create 3 files

1. server.js   2. nginx-configmap.yaml   3. nginx-pods.yaml

The content of each file is given below

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$ touch server.js nodejs-configmap.yaml nodejs-pod.yaml
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$ ls
nodejs-configmap.yaml  nodejs-pod.yaml  server.js
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$ nano server.js
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$ cat server.js
const http = require('http');
const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$ nano nodejs-configmap.yaml
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$ cat nodejs-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require('http');
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });

1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$ nano nodejs-pod.yaml
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$ cat nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      ports:
        - containerPort: 3000
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

Step 4: Start the minikube if it is not started and check the status by the command
> minikube start --driver=docker
> minikube status

Step 5: Now apply both the configmap and pod yaml files
> kubectl apply -f nodejs-configmap.yaml
> kubectl apply -f nodejs-pod.yaml

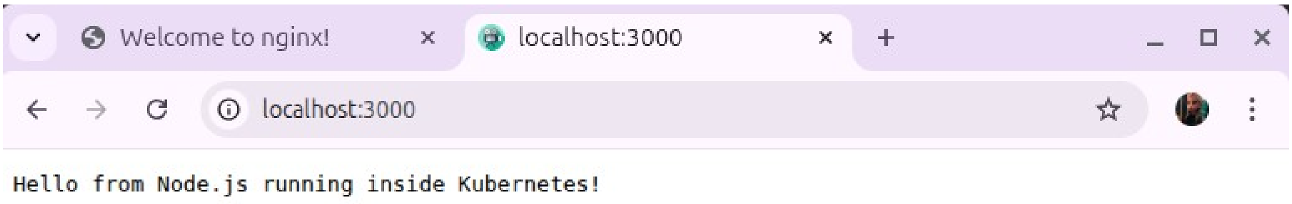
```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$ kubectl apply -f nodejs-configmap.yaml
kubectl apply -f nodejs-pod.yaml
configmap/nodejs-app-config created
pod/nodejs-pod created
```

Step 6: Check the pod status and forward the port
> kubectl get pods
> kubectl port-forward pod/nodejs-pod 3000:3000

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
nginx-deployment-6f9664446b-g9dg8   1/1     Running             0          7m35s
nginx-deployment-6f9664446b-nsmtk   1/1     Running             0          7m35s
nodejs-pod                          0/1     ContainerCreating   0          6s
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$ kubectl port-forward pod/nodejs-pod 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
```

Step 7: Open any browser and search for
> http://locahost:3000



Step 8: Now delete all the pods, services and deployments
> kubectl delete pods --all
> kubectl delete svc --all
> kubectl delete deployments –all

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$
kubectl delete pods --all
pod "nginx-deployment-6f9664446b-574wn" deleted
pod "nginx-deployment-6f9664446b-75npw" deleted
pod "nodejs-pod" deleted
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$
kubectl delete svc --all
service "kubernetes" deleted
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog8$
kubectl delete deployments --all
No resources found
```